Algorithm 1 f-DAgger

Require: Expert π_E, clean data D_{clean}, DAgger rounds K
Ensure: Learned policy π^*
Stage 1: collect expert data with piecewise OU noise
$D_{\text{OU}} \leftarrow \emptyset$
for each expert episode **do**
  Reset environment; $z \leftarrow 0$; $T_A, T_R \leftarrow \infty$; $B \leftarrow \emptyset$
  Sample one episode scale $\alpha \sim \mathcal{U}(0.25, 1)$
  for each control step t **do**
    Observe o_t and query the expert: $a_t^E \leftarrow \pi_E(o_t)$
    Update the first gate-activation time T_A and first task-reach time T_R
    $z \leftarrow \rho z + \sqrt{1-\rho^2}\,(\alpha\sigma \odot \xi)$, where $\xi \sim \mathcal{N}(0, I)$
    Apply the piecewise rule below to obtain n_t from z
    Save (o_t, a_t^E) in B ▷ clean expert label
    Execute $\text{clip}(\text{clip}(a_t^E) + n_t)$
  end for
  Add B to D_{OU} only if its sustained-hold check passes
end for
$D \leftarrow D_{\text{clean}} \uplus D_{\text{OU}}$
$\pi_0 \leftarrow \text{TrainFromScratch}(D)$
Stage 2: DAgger data aggregation
for $k = 1, \ldots, K$ **do**
  Roll out student π_{k-1} with no additional OU noise
  Label every visited observation o with $\pi_E(o)$ to obtain D_{new}
  Retain task failures; discard numerical-failure and empty episodes
  $D \leftarrow D \uplus D_{\text{new}}$
  $\pi_k \leftarrow \text{TrainFromScratch}(D)$
  Evaluate π_k on the validation scenarios
end for
return the best validation policy π^* among $\pi_1, \ldots, \pi_K$

Piecewise OU noise. Write $z = [z^{\text{arm}}; z^{\text{wheel}}]$. The applied perturbation is

$$n_t = \begin{cases} [z^{\text{arm}}; z^{\text{wheel}}], & t\Delta t < T_A, \\ [z^{\text{arm}}; 0], & T_A \le t\Delta t < T_R + 1.5 \text{ s}, \\ 0, & t\Delta t \ge T_R + 1.5 \text{ s}. \end{cases}$$

T_A is the first reach-gate activation; wheel noise remains off thereafter. T_R is the first task reach: gate active, true end-effector error below 10 mm, and elapsed time greater than 8 s. An event that never occurs has time ∞.

Defaults. $\Delta t = 0.001$ s, $\tau = 0.2$ s, $\rho = e^{-\Delta t/\tau}$; σ contains seven arm scales of 0.75 N m and four wheel scales of 0.15 rad/s. All channels use independent Gaussian innovations. $K = 3$.

Collection and training. Episodes end 3 s after the first task reach or at 40 s; numerical failures interrupt collection. Sustained-hold success requires the complete 3 s post-reach window and error below 10 mm throughout its final second. DAgger uses fitting scenarios plus fresh low/near-biased development scenarios; the student executes alone ($\beta = 0$), with command limits applied. $\uplus$ denotes data concatenation. Feature extraction and action-format conversion are implicit in the observation–expert-action pairs. Training uses the implemented BC+J objective; all previous data are retained. Validation uses sustained-hold success, with ties resolved toward later rounds.